基礎分類器

Basic Classifier

By

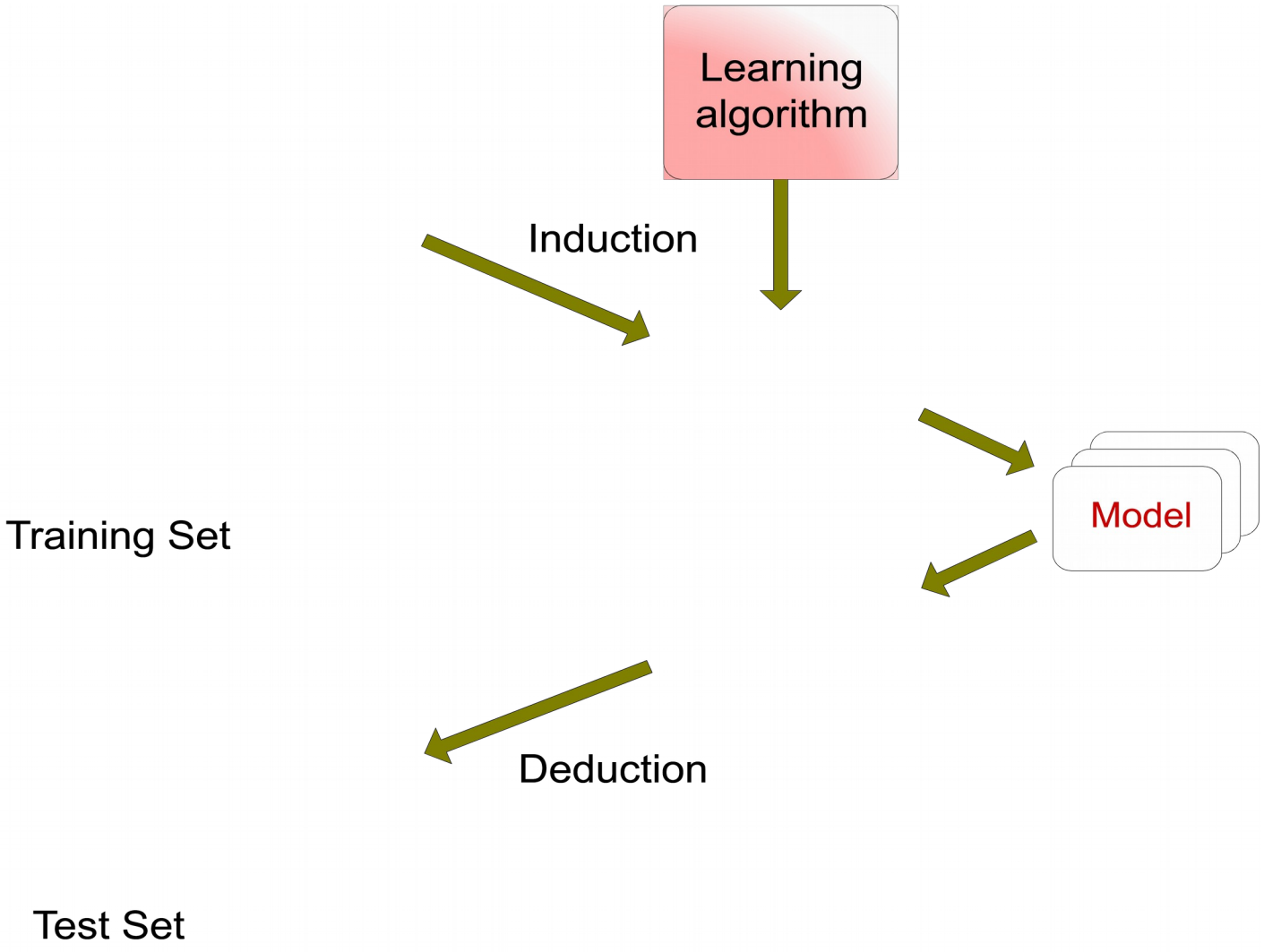
Prof. Chih-Hsuan Wang (王志軒)

甚麼是分類模型

- 給定一組資料，分類器就是針對 X(predictors) 、 Y(class- dependent variable) 變數建立預測模型
  - 通常使用訓練集建立模型、測試集驗證模型
  - 包容性高的分類器能同時接受數值型與區間型的 X
  - 功能完善的分類器還能告訴你不同 X 變數的重要性

- 實務應用：
  - 機台診斷與預測性維修
  - 顧客保留與流失預測
  - 疾病分類與良性惡性預測
  - 電商平台的推薦系統設計
  - 製程異常與瑕疵圖像分類

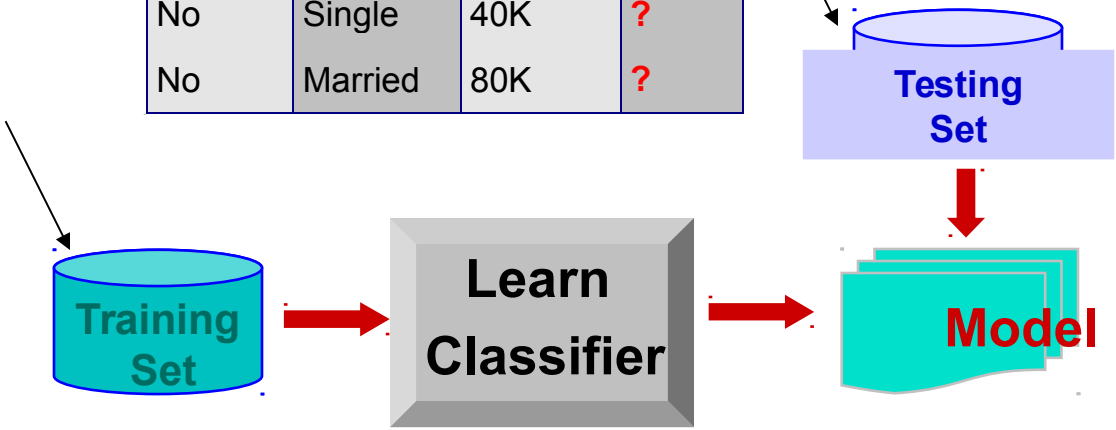
Illustrating Classification



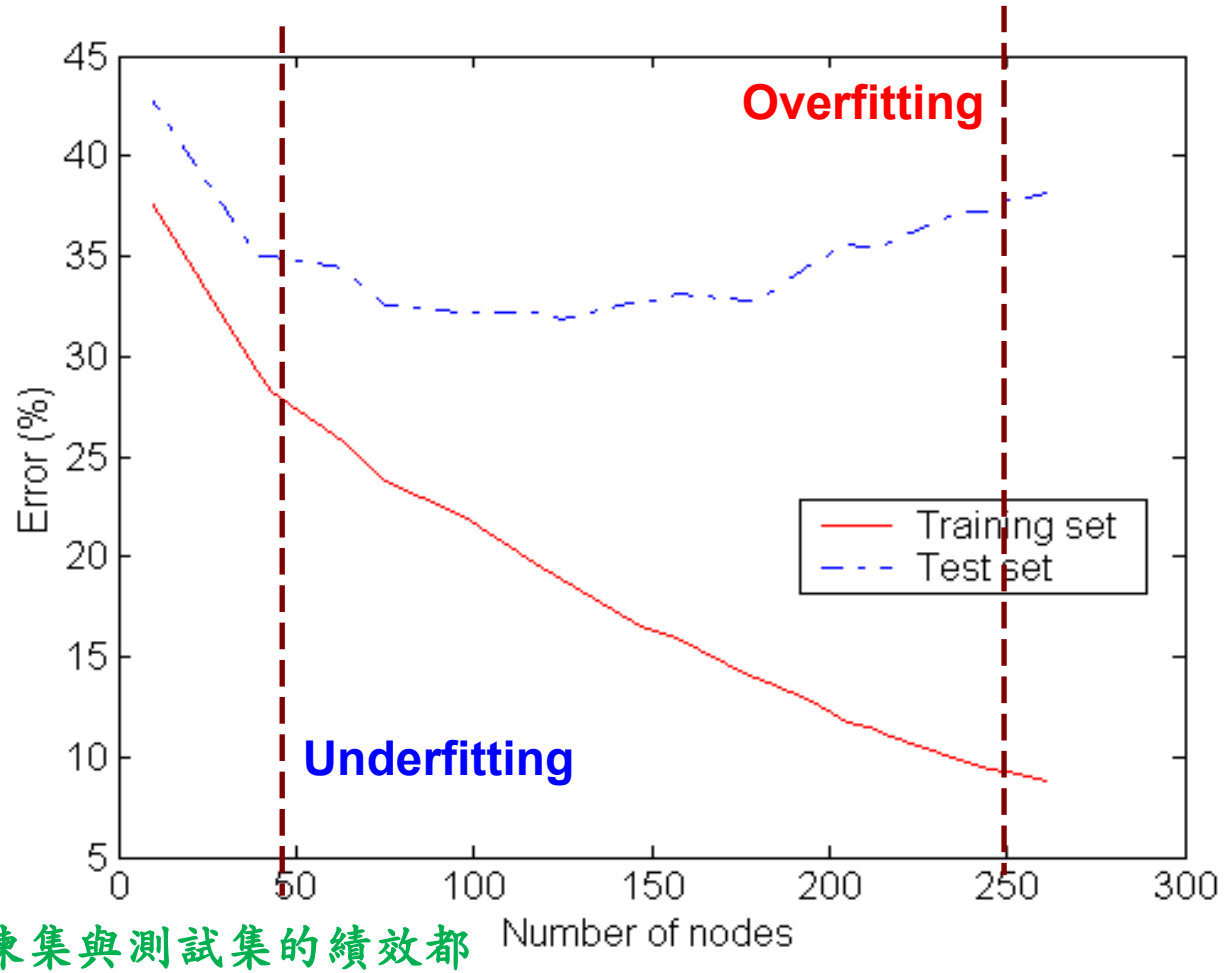
Classification as prediction

Tid	Refund (categorical)	Marital Status (categorical)	Taxable Income (continuous)	Cheat (class)
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Refund	Marital Status	Taxable Income	Cheat
No	Single	75K	?
Yes	Married	50K	?
No	Married	150K	?
Yes	Divorced	90K	?
No	Single	40K	?
No	Married	80K	?



分類器最怕過度學習



訓練集與測試集的績效差距過大，表示模型過於複雜，致使泛化能力極差

訓練集與測試集的績效都很差，表示模型過於簡單

Common Methods of Estimation

- Holdout
  - Reserve 2/3 for training and 1/3 for testing
- Cross validation (交叉驗證)
  - Partition data into k disjoint subsets
  - *k*-fold: train on k-1 partitions, test on the remaining one
  - 若有 1000 筆資料 (k=10 等分)，選擇 900 筆建模，100 筆驗證；因 100 筆可以輪替 10 回，最後將 10 次績效取平均當作最終結果
- Stratified sampling (分層抽樣)
  - oversampling vs. undersampling
- Bootstrap (抽出又放回抽樣)
  - Sampling with replacement

績效評估

- Focus on the predictive capability of a model
  - Rather than how fast it takes to classify or build models, scalability, etc.
- Confusion Matrix (混淆矩陣):

	PREDICTED CLASS		
ACTUAL CLASS		Class=Yes	Class=No
	Class=Yes	a (正陽)	b (僞陰)
	Class=No	c (僞陽)	d (正陰)

a: TP (true positive)

b: FN (false negative)

c: FP (false positive)

d: TN (true negative)

Metrics for Performance Evaluation

		PREDICTED CLASS	
		Class=Yes	Class=No
ACTUAL CLASS	Class=Yes	a (TP)	b (FN)
	Class=No	c (FP)	d (TN)

- Recall (有問題的樣本確實被揪出來的比例): a/(a+b),
- Precision (判定有問題的樣本中眞正瑕疵的比例): a/(a+c)
- F measure: 2*Recall*Precision/(Recall+Precision)

$$\text{Accuracy} = \frac{a+d}{a+b+c+d} = \frac{TP+TN}{TP+TN+FP+FN}$$

Limitation of Accuracy (正確率不是唯一)

- Consider a 2-class problem
  - Number of Class 0 examples = 9990
  - Number of Class 1 examples = 10

- If model predicts everything to be class 0, accuracy is 9990/10000 = 99.9 %
  - Accuracy is misleading because model does not detect any class 1 example
- 全部猜 class 0，正確率超過 99%，是不是很荒謬？

加入成本矩陣

	PREDICTED CLASS		
ACTUAL CLASS	C(i|j)	**Class=Yes**	**Class=No**
	Class=Yes	C(Yes|Yes)	C(No|Yes)
	Class=No	C(Yes|No)	C(No|No)

C(i|j): Cost of misclassifying class j example as class i

Cost vs. Accuracy

Cost Matrix	PREDICTED CLASS		
	C(i|j)	+	-
ACTUAL CLASS	+	-1	100
	-	1	0

Model M_1	PREDICTED CLASS		
		+	-
ACTUAL CLASS	+	150	40
	-	60	250

Accuracy = 80%

Cost = 3910

Model M_2	PREDICTED CLASS		
		+	-
ACTUAL CLASS	+	250	45
	-	5	200

Accuracy = 90%

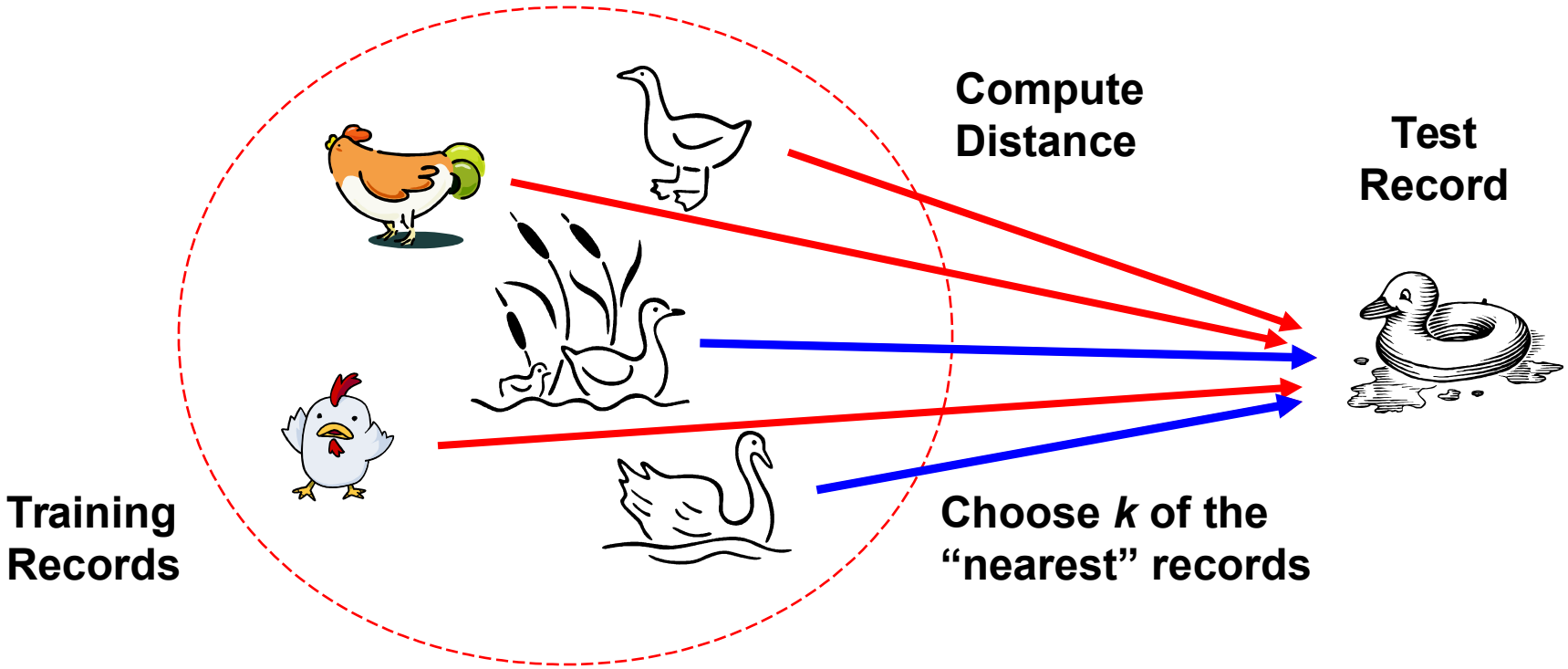
Cost = 4255

最近鄰居法 (K Nearest Neighbor)

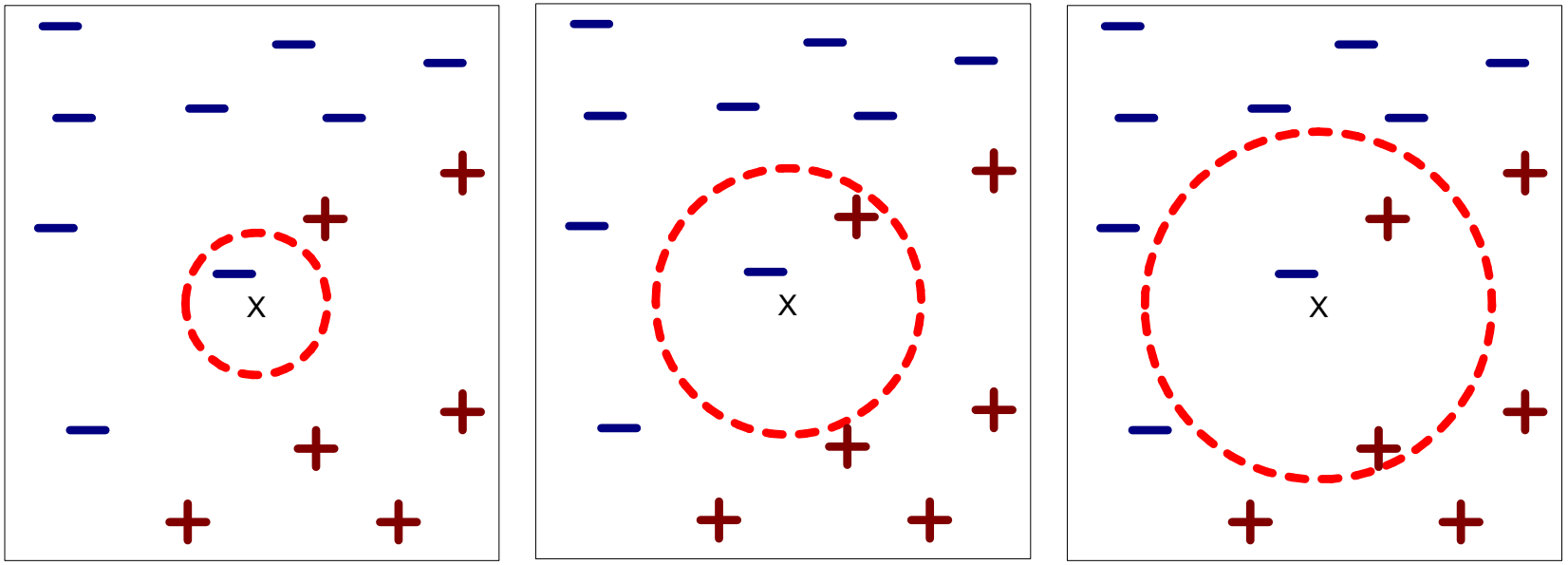
- Examples:
  - Rote-learner (以最近 K 個鄰居的類別來預測)
    - Memorizes entire training data and performs classification only if attributes of record match one of the training examples exactly
  - *K*-Nearest neighbor (近朱者赤、近墨者黑)
    - Uses *k* "closest" points (nearest neighbors) for performing classification

Nearest Neighbor Classifiers

- Basic idea:
  - If it walks like a duck, quacks like a duck, then it's probably a duck



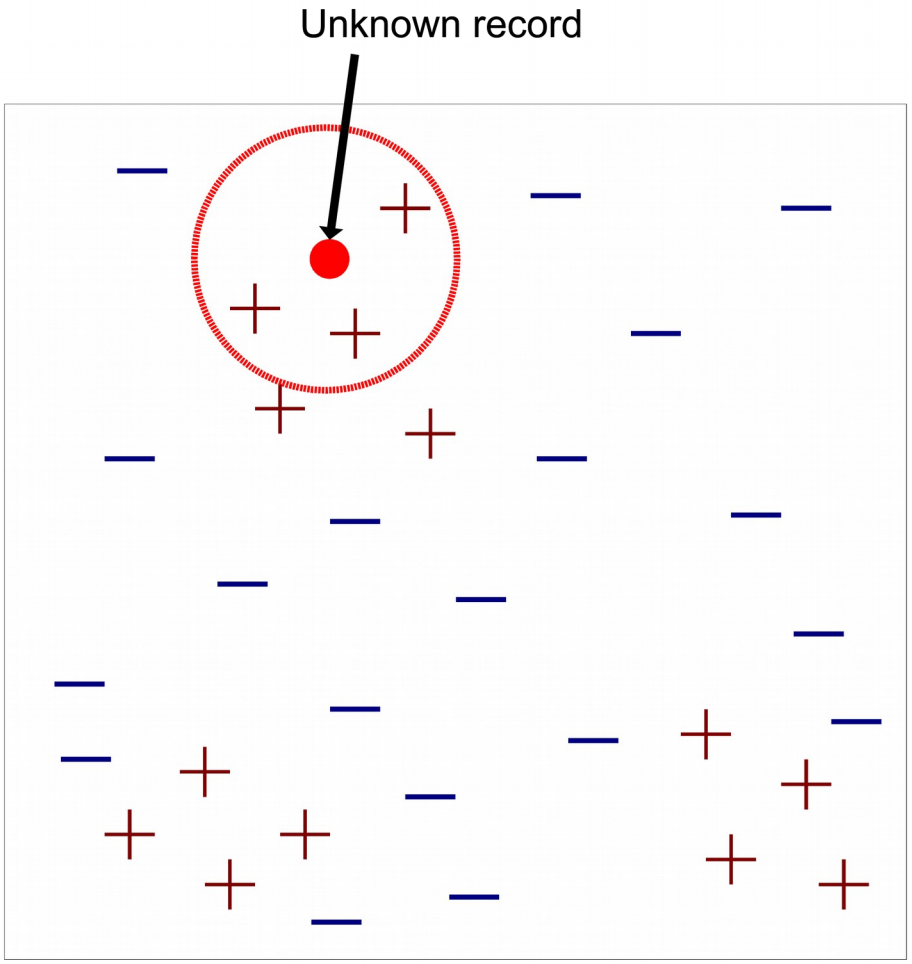
Definition of Nearest Neighbor



(a) 1-nearest neighbor (b) 2-nearest neighbor (c) 3-nearest neighbor

K-nearest neighbors of a record x are data points that have the k smallest distance to x

Nearest-Neighbor Classifiers

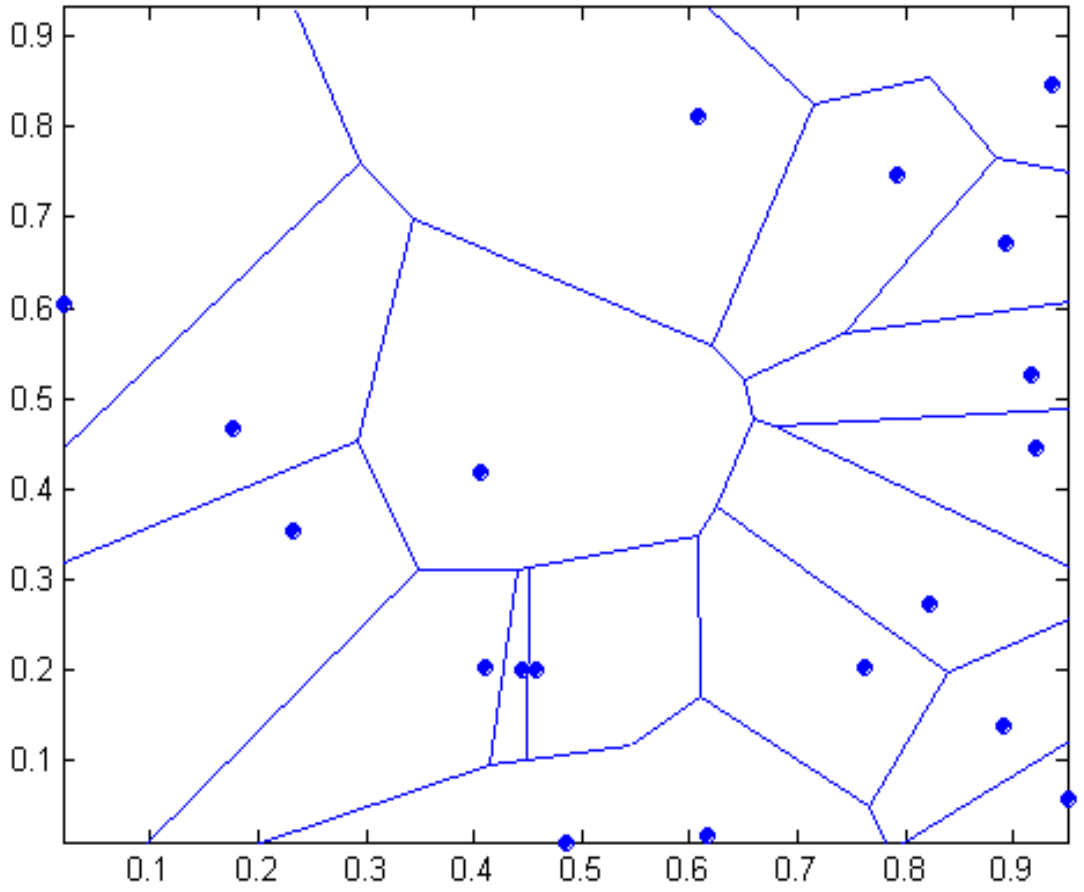


- Requires three things
  - The set of stored records
  - Distance Metric to compute distance between records
  - The value of *k*, the number of nearest neighbors to retrieve

- To classify an unknown record:
  - Compute distance to other training records
  - Identify *k* nearest neighbors
  - Use class labels of nearest neighbors to determine the class label of unknown record (e.g., by taking majority vote)

KNN (K=1 的邊界就是垂直平分線)

Voronoi Diagram



Nearest Neighbor Classification

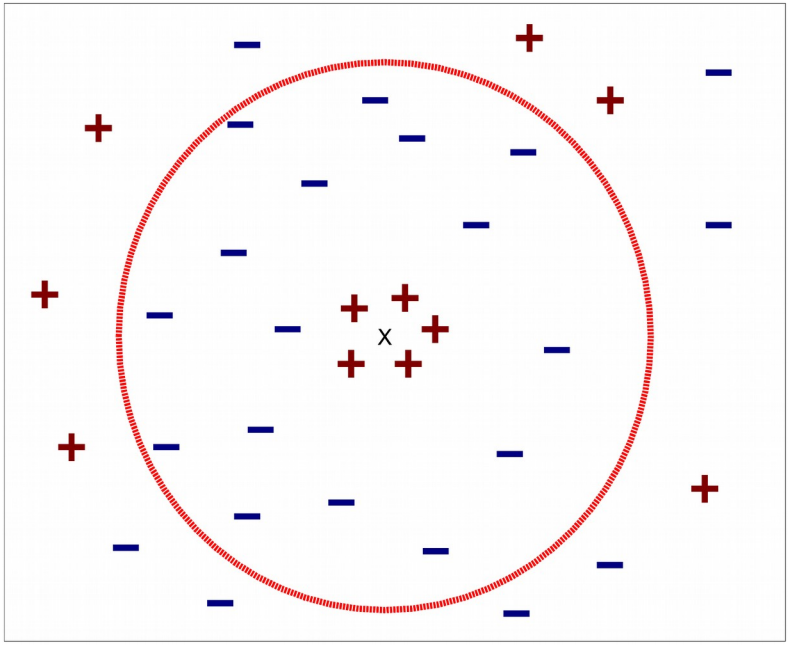
- Compute distance between two points:
  - Euclidean distance (兩樣本點間的歐式距離)

$$d(p,q) = \sqrt{\sum_i (p_i - q_i)^2}$$

- Determine the class from nearest neighbor list
  - Take majority vote of class labels among the k-nearest neighbors (樣本的 Y ，如性別、花種、故障型態等，由附近多數鄰居的 Y 是甚麼類別來決定)
  - Weigh the vote according to distance
    - weight factor, $w = 1/d^2$

Nearest Neighbor Classification…

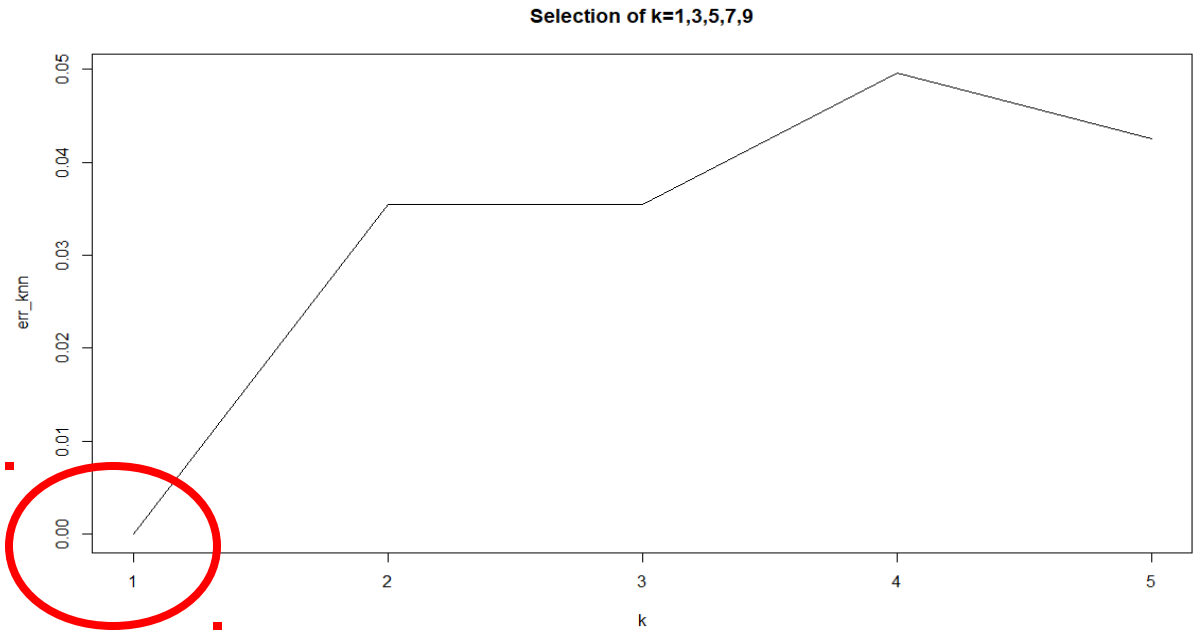
- Choosing the value of k:
  - If k is too small, sensitive to noise points (對雜訊太敏感)
  - If k is too large, neighborhood may include points from other classes (K 很大那還能稱爲鄰居嗎)



KNN for Iris dataset

Accuracy= (20+19+20)/60=98.3%, so what?

	Setosa	Versicolor	Virginica
Setosa	20		
Versicolor		19	1
Virginica			20



最近鄰居法總結

- KNN classifiers are called lazy learners (懶惰學習器)
  - It does NOT build models explicitly
  - 原理簡單，但效果不差
  - KNN 需計算歐式距離，所以 X 變數需爲數值型
  - 不同單位的 X 變數，先進行正規化 ((X-Xmin)/(Xmax-Xmin))，避免單位對分類造成困擾
  - KNN 無法指出哪個 X 變數對分類較重要
  - KNN 不是只有分類，還能進行迴歸，原理是取最近 K 個鄰居的 Y 值平均後當作預測值

Naïve Bayes Classifier (NBC: 天眞貝氏法)

- 源自於大名鼎鼎的貝氏定理
- Conditional Probability:

  A 表血便、 C 表大腸癌

$$P(C \mid A) = \frac{P(A,C)}{P(A)}$$

$$P(C \mid A) = \frac{P(A \mid C)P(C)}{P(A)} \qquad P(A \mid C) = \frac{P(A,C)}{P(C)}$$

- Bayes theorem ：看到一個血便的人，其患有大腸癌的機率 (C given A) 可從得到大腸癌會產生血便的可能性 (A given C) 乘上大腸癌的機率 (P(C)) 來估計

貝氏定理 (Bayes Theorem)

- Given:
  - A doctor knows that meningitis causes stiff neck 50% of the time
  - Prior probability of any patient having meningitis is 1/50,000
  - Prior probability of any patient having stiff neck is 1/20

- If a patient has stiff neck- 頸部痠痛 (S), what's the probability he/she has meningitis- 腦膜炎 (M): from 0.00002 to 0.0002 ，機率整整提升了 10 倍？

$$P(M|S) = \frac{P(S|M)P(M)}{P(S)} = \frac{0.5 \times 1/50000}{1/20} = 0.0002$$

貝氏分類器 (Bayesian Classifiers)

- Given a record with attributes ($A_1, A_2,\ldots,A_n$)
  - Goal is to predict class C (如由身高、體重、腰圍來預測性別)
- Can we estimate P(C| $A_1, A_2,\ldots,A_n$) directly from data?

$$P(C \mid A_1 A_2 \ldots A_n) = \frac{P(A_1 A_2 \ldots A_n \mid C) P(C)}{P(A_1 A_2 \ldots A_n)}$$

  - Choose value of C (男性或女性) that maximizes P(C | $A_1, A_2, \ldots, A_n$)
  - Equivalent to choosing value of C that maximizes P($A_1, A_2, \ldots, A_n$|C) P(C)- 到底是男生還是女生容易出現如此的身高、體重、腰圍的組合？

Naïve Bayes 假設所有的屬性彼此獨立

- Assume independence among attributes A_i when class is given:
  - $P(A_1, A_2, \ldots, A_n | C) = P(A_1 | C_j)\, P(A_2 | C_j) \ldots P(A_n | C_j)$
  - Can estimate $P(A_i | C_j)$ for all A_i and C_j.
  - New point is classified to C_j if $P(C_j) \prod P(A_i | C_j)$ is maximal ($\prod P(A_i | C_j) = P(A_1 | C_j)\, P(A_2 | C_j) \ldots P(A_n | C_j)$).

當屬性是離散型

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- Class: $P(C) = N_c/N$
  - e.g., P(No) = 7/10,
    P(Yes) = 3/10
- For discrete attributes:

$$P(A_i \mid C_k) = |A_{ik}| / N_{ck}$$

where $|A_{ik}|$ is number of instances having attribute A_i and belongs to class C_k

  - Examples:

P(Status=Married|No) = 4/7
P(Refund=Yes|Yes)=0/3

當屬性是數值型

Tid	Refund	Marital Status	Taxable Income	Evade
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

- Normal distribution:

$$P(A_i \mid c_j) = \frac{1}{\sqrt{2\pi\sigma_{ij}^2}} e^{-\frac{(A_i - \mu_{ij})^2}{2\sigma_{ij}^2}}$$

  - One for each (A_i, c_i) pair

- For (Income, Class=No):
  - If Class=No
    - sample mean = 110
    - sample variance = 2975

$$P(Income = 120 \mid No) = \frac{1}{\sqrt{2\pi}(54.54)} e^{-\frac{(120-110)^2}{2(2975)}} = 0.0072$$

給定一筆資料進行預測

$$X = (\text{Refund} = \text{No}, \text{Married}, \text{Income} = 120\text{K})$$

naive Bayes Classifier:

P(Refund=Yes|No) = 3/7
P(Refund=No|No) = 4/7
P(Refund=Yes|Yes) = 0
P(Refund=No|Yes) = 1
P(Marital Status=Single|No) = 2/7
P(Marital Status=Divorced|No)=1/7
P(Marital Status=Married|No) = 4/7
P(Marital Status=Single|Yes) = 2/7
P(Marital Status=Divorced|Yes)=1/7
P(Marital Status=Married|Yes) = 0

For taxable income:
If class=No: sample mean=110
sample variance=2975
If class=Yes: sample mean=90
sample variance=25

- P(X|Class=No) = P(Refund=No| Class=No)
  × P(Married| Class=No)
  × P(Income=120K| Class=No)
  = 4/7 × 4/7 × 0.0072 = 0.0024

- P(X|Class=Yes) = P(Refund=No| Class=Yes)
  × P(Married| Class=Yes)
  × P(Income=120K| lass=Yes)
  = 1 × 0 × 1.2 × 10^{-9} = 0

Since P(X|No)P(No) > P(X|Yes)P(Yes)

Therefore P(No|X) > P(Yes|X)

=> Class = No (判定 Y 爲不會逃稅)

Naïve Bayes 物種分類

Name	Give Birth	Can Fly	Live in Water	Have Legs	Class
human	yes	no	no	yes	mammals
python	no	no	no	no	non-mammals
salmon	no	no	yes	no	non-mammals
whale	yes	no	yes	no	mammals
frog	no	no	sometimes	yes	non-mammals
komodo	no	no	no	yes	non-mammals
bat	yes	yes	no	yes	mammals
pigeon	no	yes	no	yes	non-mammals
cat	yes	no	no	yes	mammals
leopard shark	yes	no	yes	no	non-mammals
turtle	no	no	sometimes	yes	non-mammals
penguin	no	no	sometimes	yes	non-mammals
porcupine	yes	no	no	yes	mammals
eel	no	no	yes	no	non-mammals
salamander	no	no	sometimes	yes	non-mammals
gila monster	no	no	no	yes	non-mammals
platypus	no	no	no	yes	mammals
owl	no	yes	no	yes	non-mammals
dolphin	yes	no	yes	no	mammals
eagle	no	yes	no	yes	non-mammals

Give Birth	Can Fly	Live in Water	Have Legs	Class
yes	no	yes	no	?

A: attributes

M: mammals

N: non-mammals

$$P(A|M) = \frac{6}{7} \times \frac{6}{7} \times \frac{2}{7} \times \frac{2}{7} = 0.06$$

$$P(A|N) = \frac{1}{13} \times \frac{10}{13} \times \frac{3}{13} \times \frac{4}{13} = 0.0042$$

$$P(A|M)P(M) = 0.06 \times \frac{7}{20} = 0.021$$

$$P(A|N)P(N) = 0.004 \times \frac{13}{20} = 0.0027$$

P(A|M)P(M) > P(A|N)P(N)

=> Mammals (哺乳類)

Naïve Bayes Classifier (總結)

- 對雜訊與不相干的屬性不敏感 (較不易受干擾)
- 能夠同時接受數值型或間斷型的 X 變數
- 只能進行分類 (Y 類別)，無法進行迴歸 (Y 數值)
- 依舊無法告訴使用者哪個 X 變數重要
- 假設所有 X 變數彼此獨立，若要放寬限制，需改用 Bayesian Belief Networks (貝氏信賴網路)

NBC for Iris (機率輸出是其特色)

	Setosa	Versicolor	Virginica
Setosa	20		
Versicolor		20	0
Virginica		2	18

Accuracy= (20+20+18)/60=96.7%

透露 Versicolor 與 Virginica 容易混淆

	Setosa	Versicolor	Virginica
Sample 1	1	1.18x10^-16	1.18x10^-22
Sample 51	10^-50	0.999	1.61x10-5
Sample 101	3.74x10^-135	0.102	0.877

NBC for Titanic (類別 / 數值 X 變數)

Discrete	Female	Male	Class 1	Class 2	Class3
No	0.179	0.821	0.139	0.199	0.621
Yes	0.65	0.35	0.416	0.23	0.354

以 X: Gender, Class, Age, Fare 來預測 Y: survival (No/Yes)

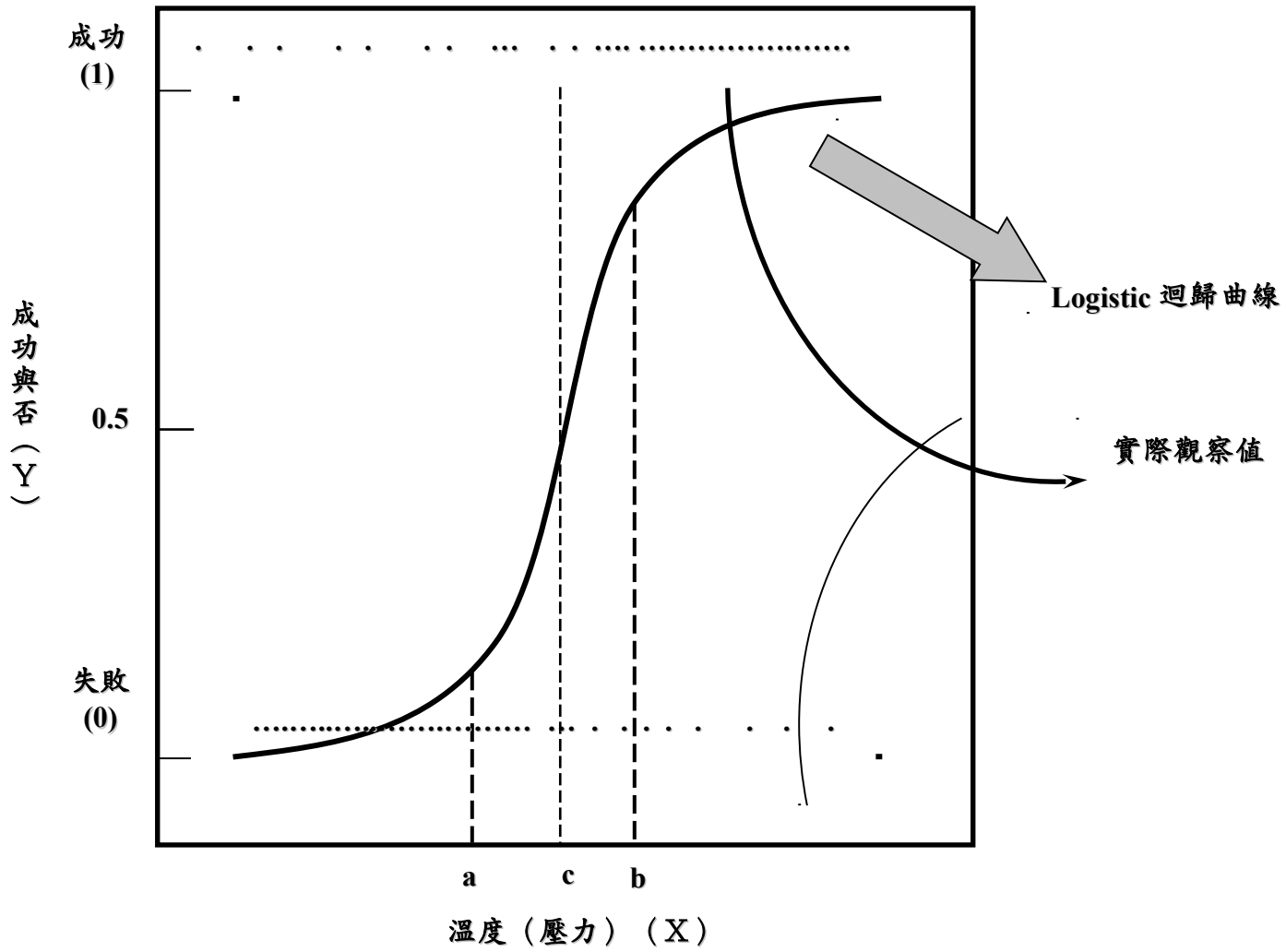
Numerical	Age		Fare	
No	29.63	13.5	22.99	34.18
Yes	28.89	14.22	50.45	69.71

羅吉斯迴歸 (Logistic Regression)

- Logistic 迴歸是以迴歸模式來建立一個依變數 Y 與一個或多個預測變數 X 之間的分類關係，Logistic 迴歸所探討結果的依變數是離散型，特別是其分類只有二元（如“正常與異常”、“良性與惡性”、“存活與死亡”）或少數幾類時（多元羅吉斯）。

- Logistic 迴歸的目的是在於建立一個最精簡和最能配適 (fit) 的分析結果，Logistic 迴歸不需要假設分配類型，在 Logistic 分配中，自變數 (X) 對於依變數 (Y) 之影響方式是以指數的方式來變動，即 $y=e^{f(x)}$ 。

Logistic 概念的圖示



Logistic 迴歸模式

- 在 Logistic 迴歸分析中，依變數必須爲類別變數，而自變數可以是類別變數 (需使用 dummy variable 技巧) ，也可以是數值變數；如地區變數以 (1,0,0) 表示北區、(0,1,0) 表示中區、(0,0,1) 表示南區、 (0,0,0) 表示東區。

令 p 表示某種事件成功的機率，它受因素 X 的影響，即 p 與 x 之關係如下：

$$P = \frac{e^{f(x)}}{1 + e^{f(x)}}$$

$$\ln \frac{p}{1-p} = f(x) = \beta_0 + \beta_1 X + \beta_2 X_2 + \cdots\cdots + \beta_k X_k$$

P/(1-P): odds ratio (成功機率除以失敗機率謂之勝算比)

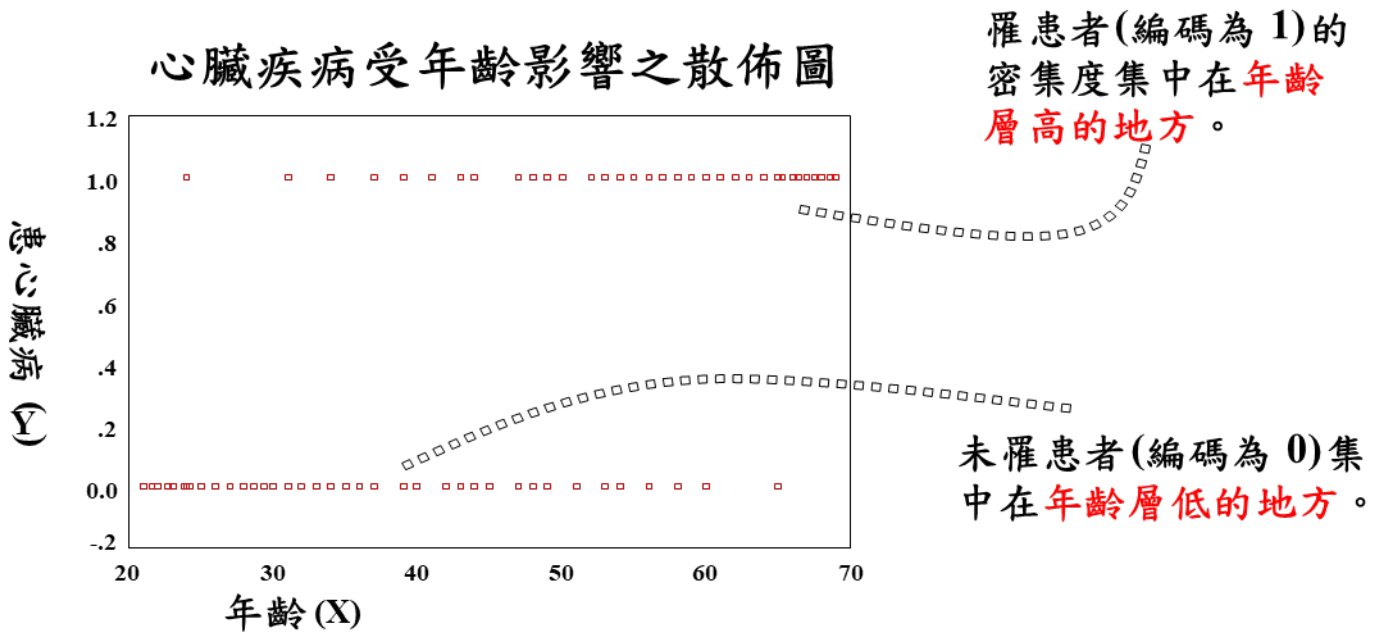
Logistic 迴歸分析的評估 (單變數)

下表是收集 125 位年齡從 21 歲到 69 歲成年人的資料，研究年齡X與是否患心臟病 (Y=0 表未患心臟病， Y=1 表患心臟病) 的關係；並依下表繪出Y對X的散佈情形，如下圖 所示。

年齡 (X) 與罹患心臟病 (Y) 之關係表

X	Y	X	Y	X	Y	X	Y	X	Y	X	Y	X	Y	X	Y	X	Y
21	0	26	0	32	0	36	1	41	1	45	0	54	0	57	1	61	1	67	1
21	0	27	0	33	0	36	0	42	0	45	0	54	1	58	0	62	0	67	1
22	0	27	1	33	0	36	0	42	0	46	0	54	0	58	0	62	0	68	1
22	0	28	0	33	0	37	1	42	1	47	0	55	1	58	1	62	1	68	1
22	0	28	0	34	0	37	0	43	0	47	1	56	0	58	1	63	1	69	1
23	0	29	1	34	1	38	1	43	1	47	0	56	0	59	1	64	1		
24	0	30	0	34	1	39	0	43	0	47	0	56	1	59	0	64	1		
24	1	30	0	34	0	39	1	43	1	48	1	57	0	60	0	65	1		
25	0	30	0	35	0	39	0	44	0	48	1	57	1	60	0	65	1		
25	0	31	0	35	0	40	0	44	0	48	0	57	1	60	1	65	0		
26	0	31	0	35	0	40	1	44	1	48	1	57	0	60	1	66	1		
26	0	31	1	36	0	40	0	45	0	49	1	57	1	61	1	66	1		

Logistic 迴歸分析的圖示



Y與X的相關係數爲 **0.4641**。　（使用直線迴歸模式）

Y對X的簡單迴歸式爲：$Y = -0.268 + 0.01524\,X$

$R^2 = 0.168$

由於主題屬於 0、1 之資料類別，所以由 R^2 僅達 0.168 可看出，本範例不適合使用線性迴歸模式

Logistic 迴歸分析的評估

當以 Logistic 迴歸作爲考量，則設年齡 X 與患心臟病機率的關係式爲：

$$P = \pi(x) = \frac{e^{b_0 + b_1 X}}{1 + e^{b_0 + b_1 X}}$$

經過 Logistic 轉換： $g(x) = \ln \frac{\pi(x)}{1 - \pi(x)} = b_0 + b_1 X$

則此時與 X 呈線性關係，即可進行 Logistic 迴歸分析。最後，數據得到以下的方程式：

$$\ln \frac{p}{1-p} = f(x) = -3.6064 + 0.0712 X$$

Logistic 迴歸的應用

- 在 Logistic 迴歸式，假設病患的年齡爲 31 歲，以 X=31 代入 Logistic 迴歸式中，可得知其患有心臟病的機率爲 19.79%:

$$\pi(31) = \frac{e^{-0.36064+0.0712*(31)}}{1+e^{-0.36064+0.0712*(31)}} = 0.19794309 < 0.5$$

- 如果想要知道年齡要超過多少歲，心臟病的機率會大於 50% 。則可使用下列式子獲得年齡要超過 50.65 歲 :

$$\pi(X) = \frac{e^{-0.36064+0.0712*X}}{1+e^{-0.36064+0.0712*X}} > 0.5$$

$\Rightarrow e^{-0.36064+0.0712*X} > 0.5*(1+e^{-0.36064+0.0712*X})$

$\Rightarrow e^{-0.36064+0.0712*X} > 1$

兩邊同取ln後可得

$\Rightarrow -0.36064+0.0712*X>0$

$\Rightarrow X > 50.65$

Logistic Regression (總結)

- 對雜訊與不相干的屬性較敏感而易受干擾
- 能夠同時接受數值型或間斷型的 X 變數
- 只能進行分類 (Y 類別)，無法進行迴歸 (Y 數值)
- 可以告訴使用者哪個 X 變數重要 (迴歸係數顯著)
- 最大的優點是呈現不同預測類別的相對機率 (如計算 healthy, risky, sick 三類的相對機率)

Logit for gender (添予預測機率，並指出顯著變數)

	Estimate	Std. Error	P value<0.05
Intercept	-137.448	45.559	0.00124
Height	0.422	0.157	0.00717
Weight	-0.095	0.109	0.38152
Waist	2.268	0.725	0.0045

為什麼體重在預測性別時不顯著 ??

	Female	Male
Female	51	3
Male	4	83

Logit 最棒之處在於算出類別機率，假設身高、體重、腰圍資料如下來預測性別 (female, male) ：
(175, 69, 31.5) → (5%, 95%)
(165, 50, 29) → (83%, 17%)

ROC (Receiver Operating Characteristic)

- Developed in signal detection theory to analyze noisy signals and characterize the trade-off between positive hits (精準度) and false alarms (靈敏度)

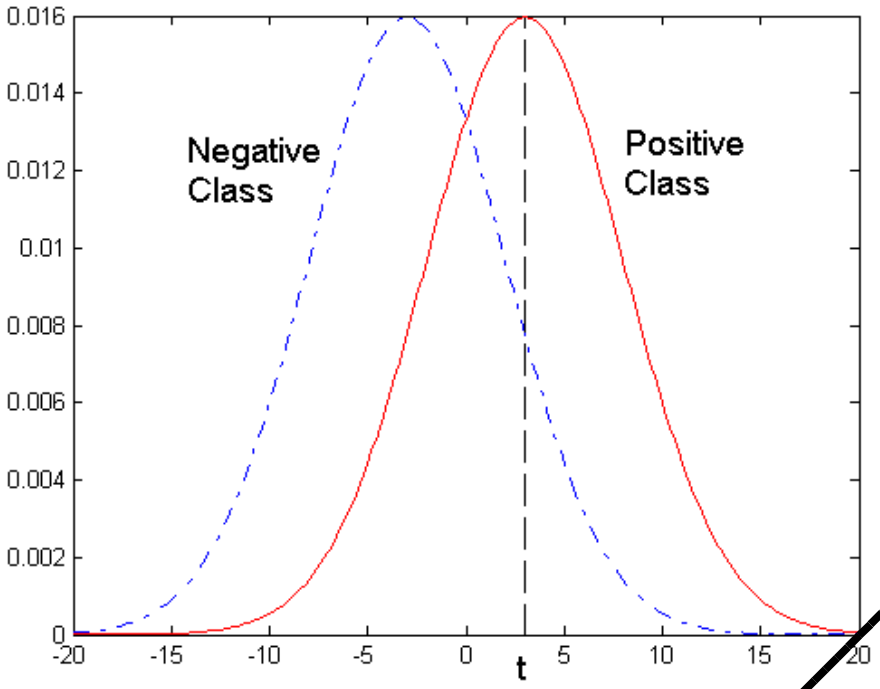
- 如何評估分類器績效 : ROC curve plots TP (on the y-axis) against FP (on the x-axis)

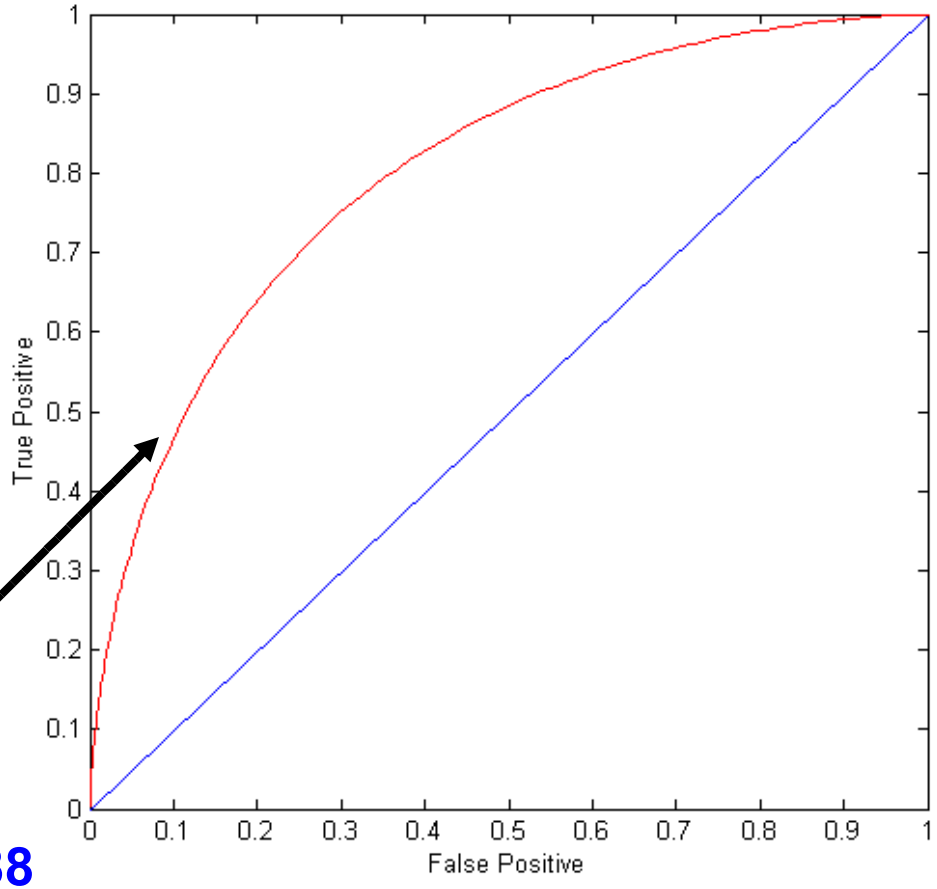
- 統計學上的型一錯誤 (false alarm- 沒病誤判成有病) 與型二錯誤 (miss alarm- 有病誤判成沒病), 哪種情況嚴重？

ROC Curve

- 1-dimensional data set containing 2 classes (positive and negative)

- any points located at x > *t* is classified as positive





At threshold t (臨界值):

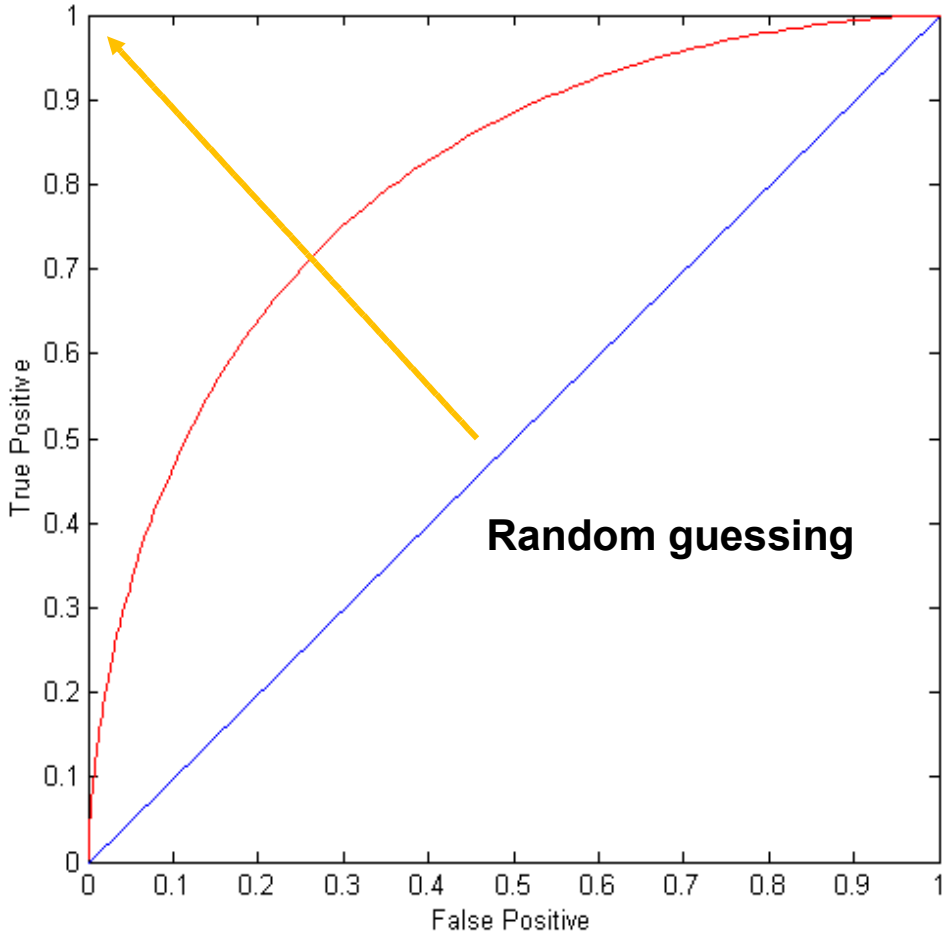
TP=0.5, FN=0.5, FP=0.12, TN=0.88

ROC Curve

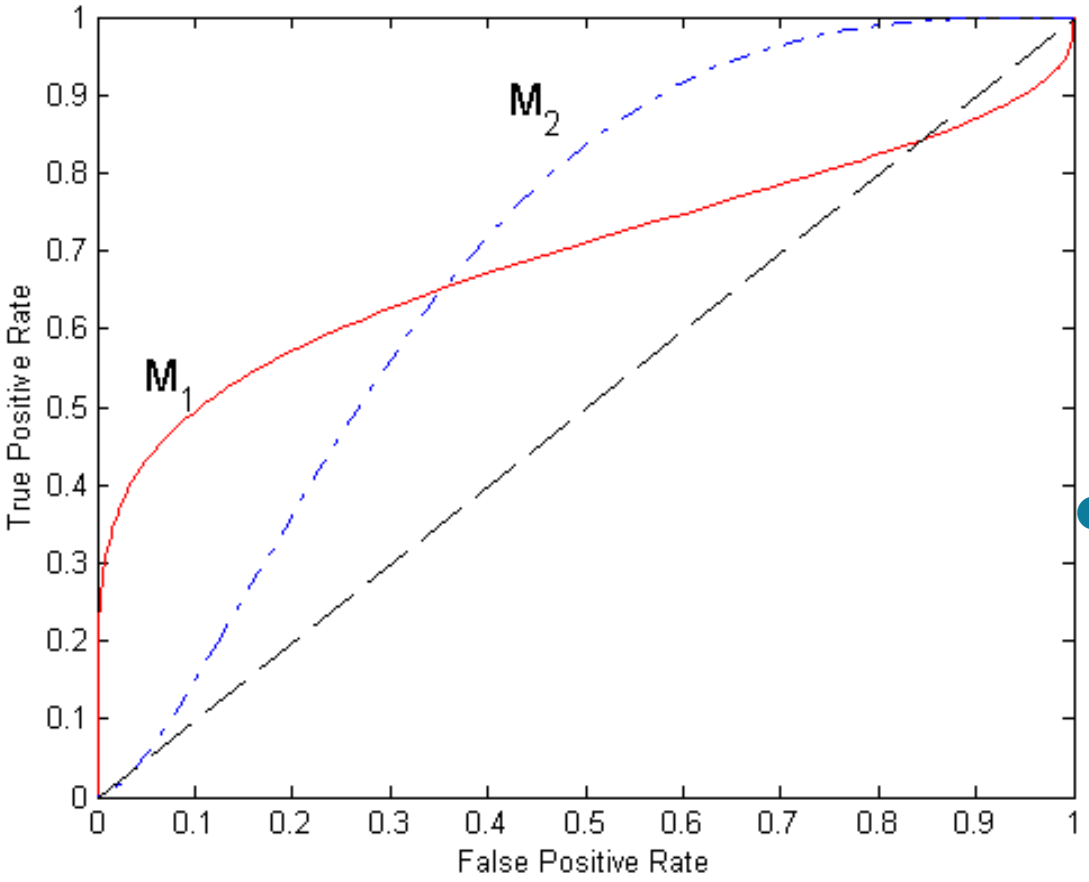
(TP,FP):

- (0,0): declare everything to be negative class
- (1,1): declare everything to be positive class
- (1,0): ideal (理想點)

- Diagonal line:
  - Random guessing
  - Below diagonal line:
    - prediction is opposite of the true class



Using ROC for Model Comparison



- No model consistently outperform the other
  - M_1 is better for small FPR
  - M_2 is better for large FPR
- Area Under the ROC curve (面積越大越好)
  - Ideal: (正方形面積)
    - Area = 1
  - Random guess:
    - Area = 0.5 (三角形面積)